

Validacao visual das predicoes - MNIST

Imagem original, mapas de oclusao e diferenca entre NEAT puro e NEAT + SGD.

Amostras no PDF: 30

Imagens na metrica global: 10000 (full_loaded_test_set)

Metodo de atribuicao: occlusion_absolute_predicted_logit_change_v2

Dataset: MNIST

Resolucao: 26x26

Retreino dos pesos: SGD

Inicializacao pos-NEAT: random_reset_preserving_neat_topology

Variante NEAT escolhida: neat_mnist_seeded_prototypes_full

Baseline: 84.7%

NEAT puro: 82.9%

Topologia NEAT + SGD: 77.5%

Conexoes ativas: 200

NEAT puro vs NEAT + SGD (mesma arquitetura):

Divergencia de atencao no dataset: 0.323

Divergencia quando preveem a mesma classe: 0.286 (n=8528)

Cobertura de mapas validos: 100.0%

Jaccard ponderado quando ambos acertam: 0.714 (n=7438)

Jaccard ponderado quando ambos erram: 0.658 (n=1396)

Gap de concordancia acerto - erro: +0.057

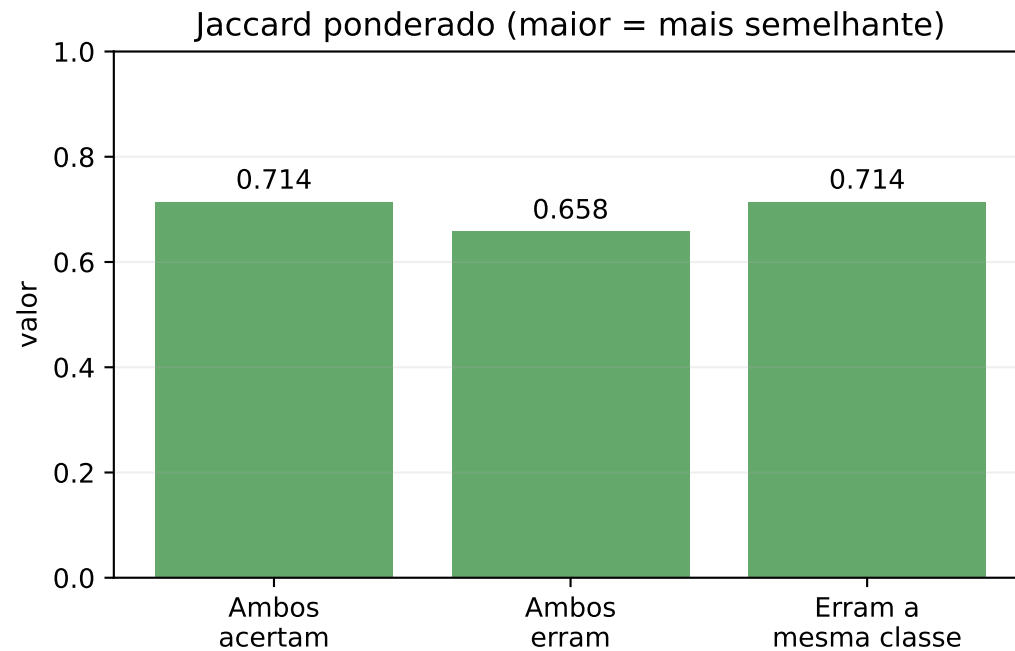
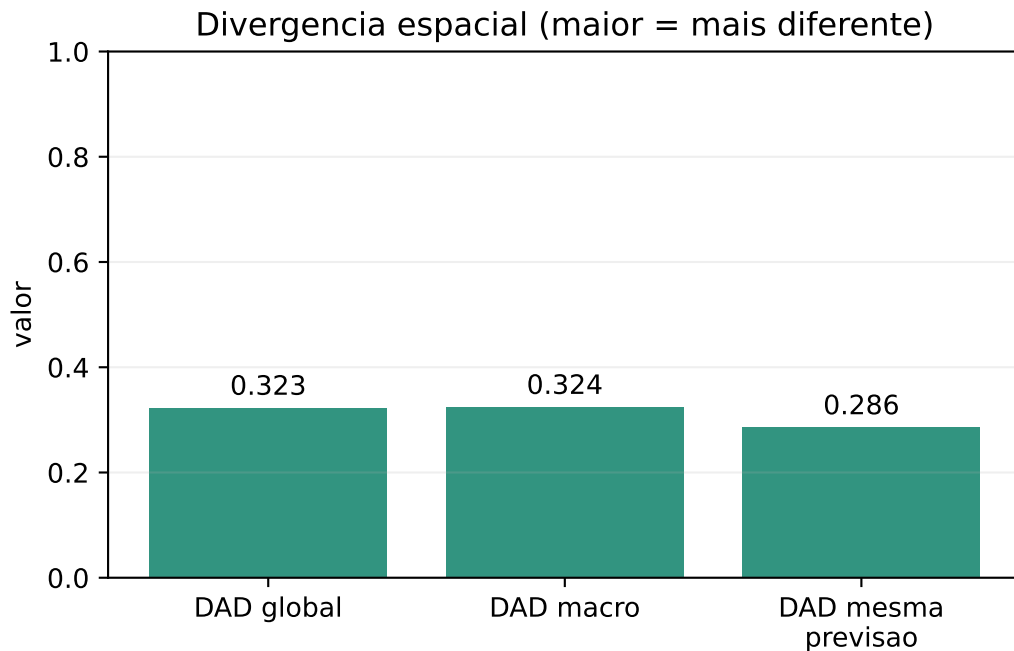
Erros com mesma classe prevista: 0.714 (n=1090)

Gap controlado acerto - erro: -0.000

Erros conjuntos com classes diferentes: 21.9%

Variante	NEAT puro	NEAT + SGD	Fitness	Conexoes
neat_mnist_features_full	37.6%	85.6%	0.801	618
neat_mnist_seeded_prototypes_full	82.9%	77.5%	1.627	200
hyperneat_mnist_prototypes_cppn	82.8%	77.9%	1.609	16

Sumarizacao quantitativa da interpretabilidade



Dataset: mnist | seed: 42 | imagens: 10000

Acuracias - baseline: 84.7% | NEAT puro: 82.9% | NEAT + SGD: 77.5%

DAD: 0.323 [IC95% 0.321, 0.325] | cobertura: 100.0%

Gap acerto-erro: +0.057 | gap controlado: -0.000

Grupos - ambos acertam: 7438 | ambos erram: 1396 | apenas um acerta: 1166

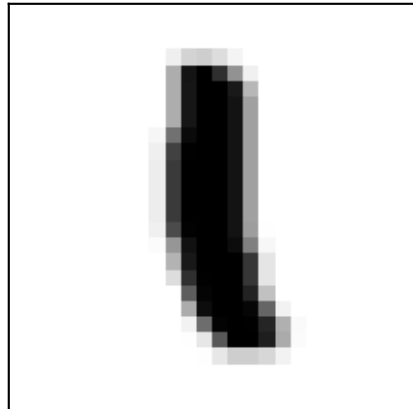
Definicoes das metricas

A DAD e uma metrica operacional deste estudo, baseada na similaridade min-max.

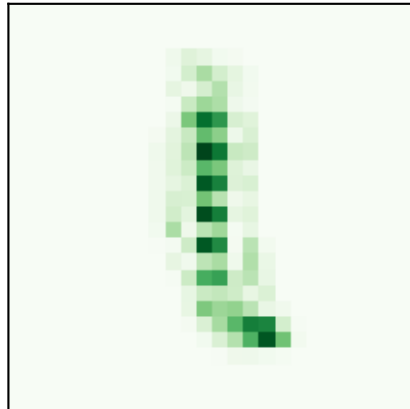
Metrica	Definicao	Faixa	Interpretacao
dataset_attention_divergence	1 menos a media do Jaccard ponderado entre mapas pareados.	0 a 1	Maior valor indica maior divergencia espacial media.
dad_macro	DAD calculada com o mesmo peso para cada classe.	0 a 1	Controla desequilibrio na quantidade de imagens por classe.
dad_same_prediction	DAD apenas quando os modelos preveem a mesma classe.	0 a 1	Compara mapas que explicam a mesma classe-alvo.
valid_map_coverage	Fracao de imagens com sinal de oclusao valido nos dois mapas.	0 a 1	Valores proximos de 1 indicam boa cobertura da sumariazacao.
both_correct_weighted_jaccard	Concordancia min-max media quando ambos acertam.	0 a 1	Maior valor indica regioes mais semelhantes nos acertos.
both_wrong_weighted_jaccard	Concordancia min-max media quando ambos erram.	0 a 1	Maior valor indica regioes mais semelhantes nos erros.
both_wrong_same_prediction_weighted_jaccard	Concordancia nos erros em que ambos preveem a mesma classe.	0 a 1	Controla a classe-alvo ao comparar os erros.
attention_agreement_gap	Jaccard medio nos acertos menos Jaccard medio nos erros.	-1 a 1	Positivo indica maior concordancia espacial nos acertos.
controlled_attention_gap	Gap usando somente erros com a mesma classe incorreta.	-1 a 1	Controla a classe-alvo explicada pelos heatmaps.
wrong_prediction_disagreement	Fracao dos erros conjuntos com classes previstas diferentes.	0 a 1	Maior valor indica erros de decisao mais distintos.

Pagina 1 - discordam - apenas NEAT puro acerta - amostra 40 - rotulo real 1
Baseline: 1 (100.0%) | NEAT puro: 1 (11.6%) | NEAT + SGD: 8 (34.2%) | similaridade NEAT/SGD: 0.78 | pixels comuns: 86.1%

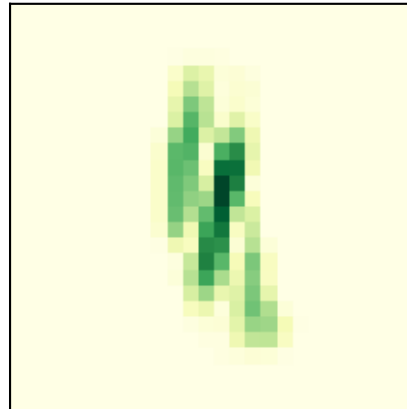
Imagem original



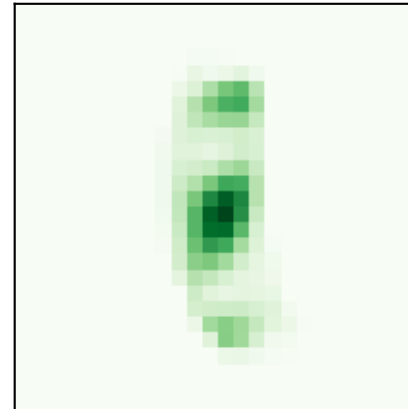
Sensibilidade: baseline



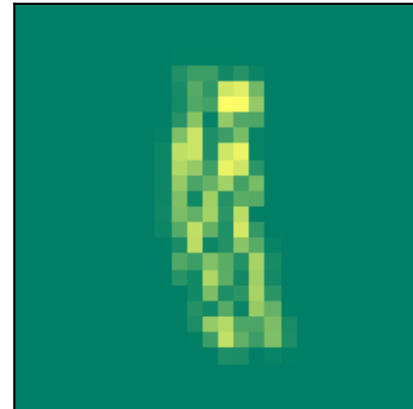
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

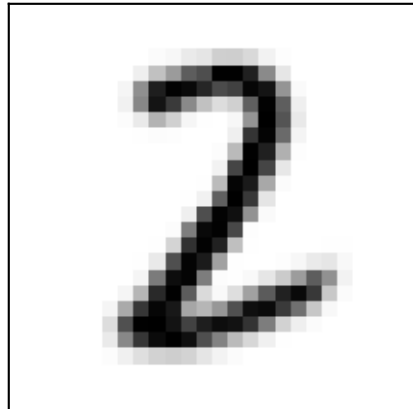


Diferença NEAT vs SGD

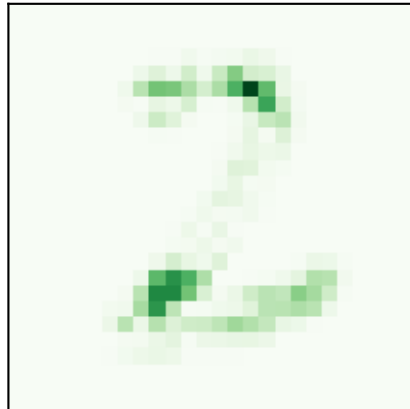


Pagina 2 - discordam - apenas NEAT puro acerta - amostra 211 - rotulo real 2
Baseline: 2 (99.8%) | NEAT puro: 2 (11.9%) | NEAT + SGD: 3 (28.0%) | similaridade NEAT/SGD: 0.86 | pixels comuns: 83.2%

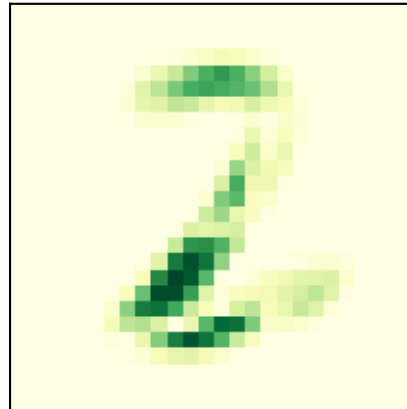
Imagem original



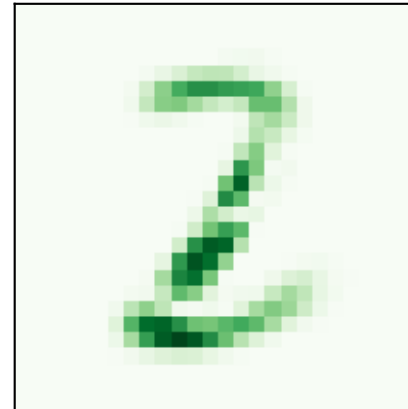
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

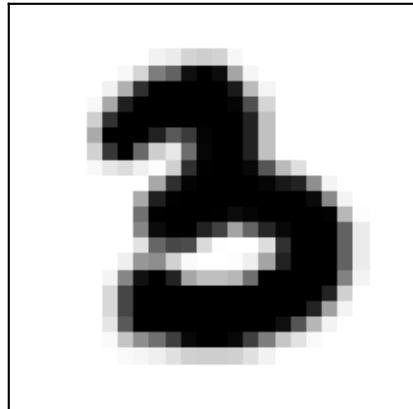


Diferença NEAT vs SGD

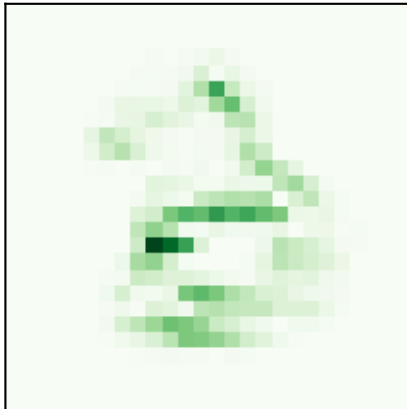


Pagina 3 - discordam - apenas NEAT puro acerta - amostra 183 - rotulo real 3
Baseline: 0 (98.5%) | NEAT puro: 3 (11.3%) | NEAT + SGD: 8 (24.7%) | similaridade NEAT/SGD: 0.76 | pixels comuns: 64.4%

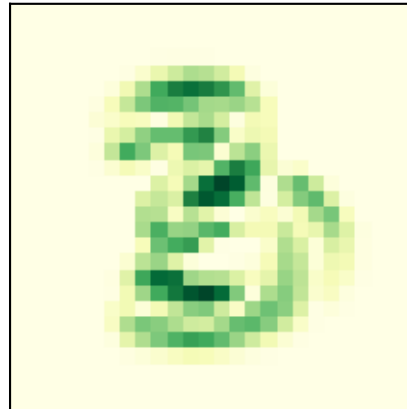
Imagem original



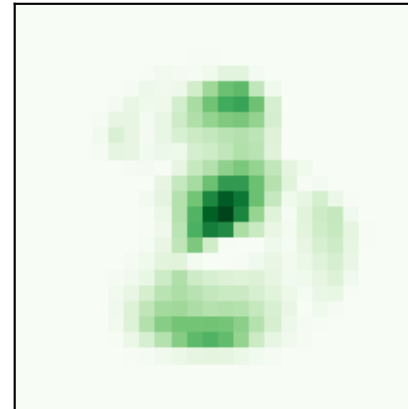
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

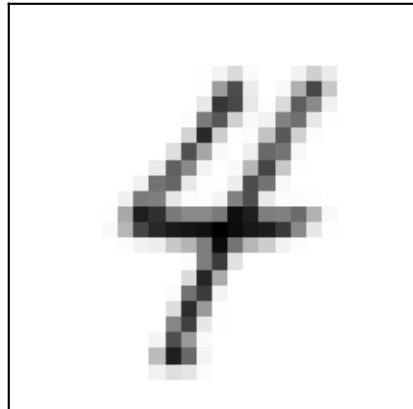


Diferença NEAT vs SGD

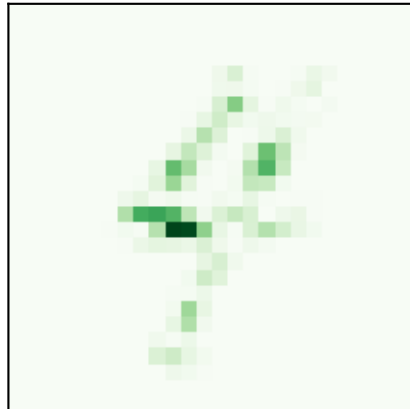


Pagina 4 - discordam - apenas NEAT puro acerta - amostra 72 - rotulo real 4
Baseline: 4 (100.0%) | NEAT puro: 4 (11.9%) | NEAT + SGD: 9 (28.0%) | similaridade NEAT/SGD: 0.83 | pixels comuns: 89.1%

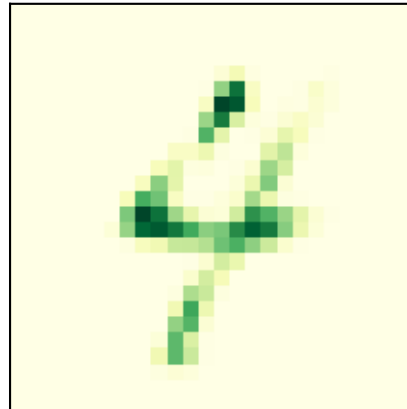
Imagem original



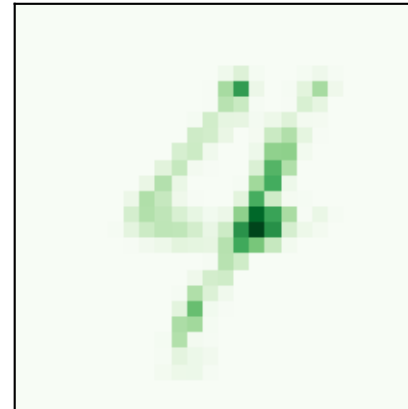
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

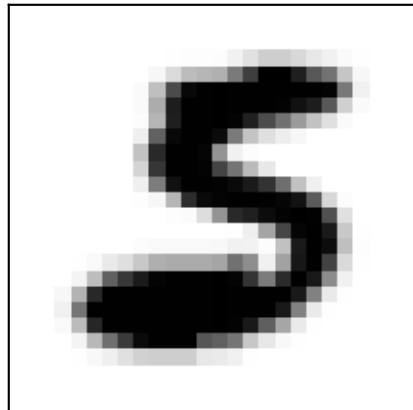


Diferença NEAT vs SGD

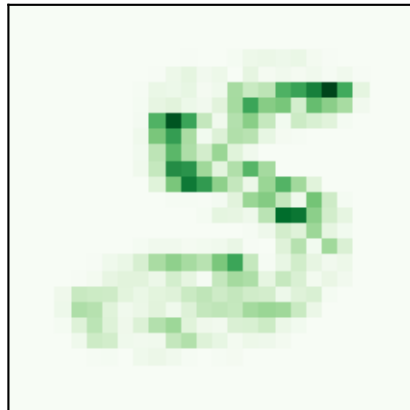


Pagina 5 - discordam - apenas NEAT puro acerta - amostra 25 - rotulo real 5
Baseline: 5 (100.0%) | NEAT puro: 5 (11.1%) | NEAT + SGD: 3 (21.7%) | similaridade NEAT/SGD: 0.73 | pixels comuns: 70.3%

Imagem original



Sensibilidade: baseline



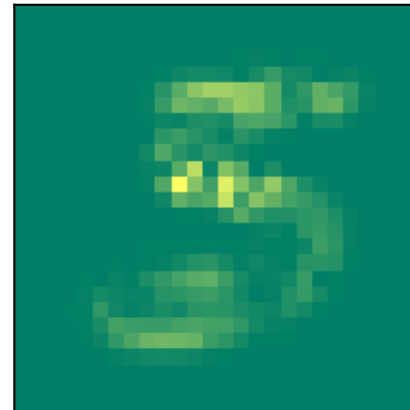
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

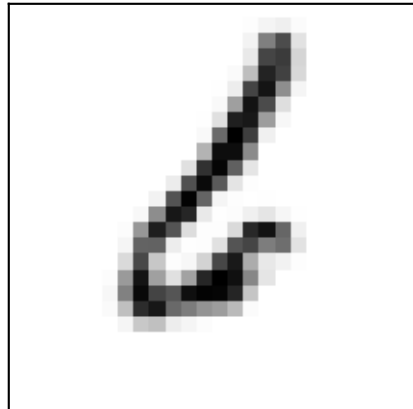


Diferença NEAT vs SGD

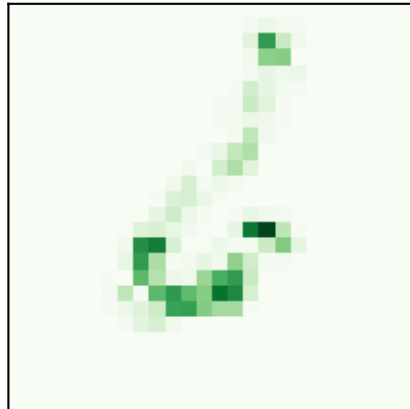


Pagina 6 - discordam - apenas NEAT puro acerta - amostra 108 - rotulo real 6
Baseline: 6 (83.6%) | NEAT puro: 6 (11.1%) | NEAT + SGD: 2 (22.0%) | similaridade NEAT/SGD: 0.75 | pixels comuns: 87.1%

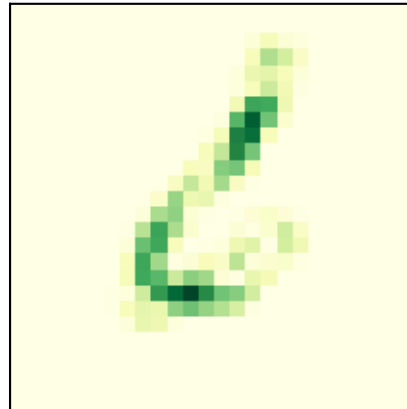
Imagem original



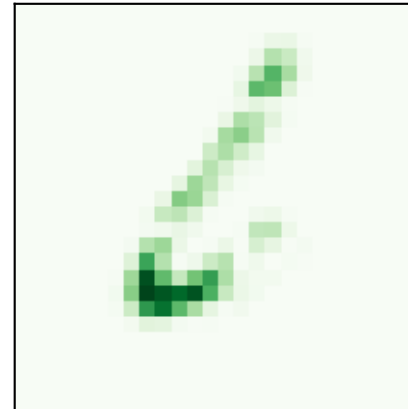
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

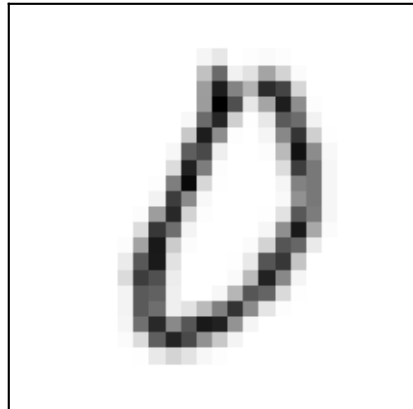


Diferença NEAT vs SGD

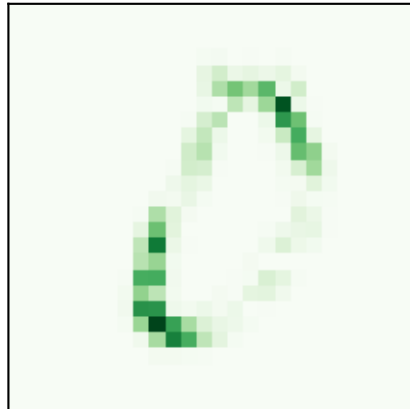


Pagina 7 - discordam - apenas NEAT + SGD acerta - amostra 66 - rotulo real 0
Baseline: 0 (100.0%) | NEAT puro: 5 (11.4%) | NEAT + SGD: 0 (27.0%) | similaridade NEAT/SGD: 0.63 | pixels comuns: 74.3%

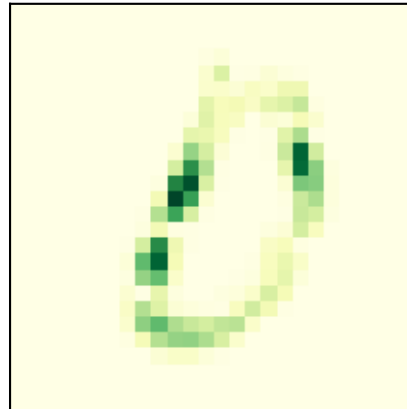
Imagem original



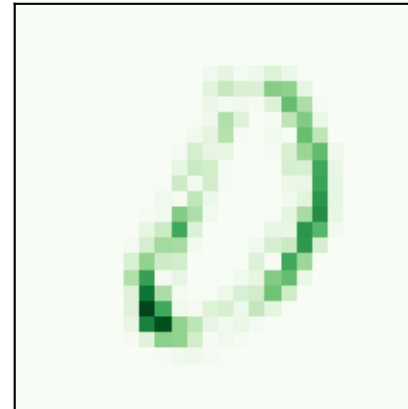
Sensibilidade: baseline



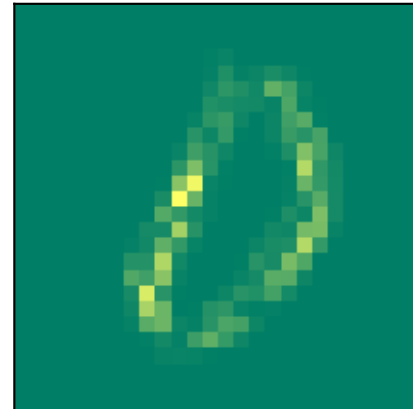
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

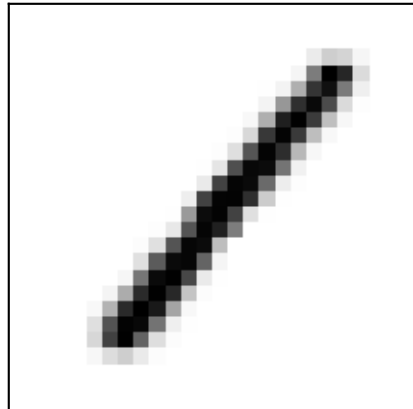


Diferença NEAT vs SGD

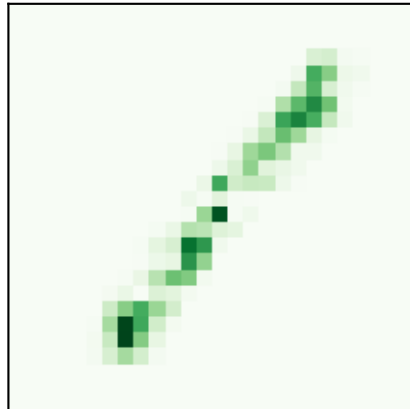


Pagina 8 - discordam - apenas NEAT + SGD acerta - amostra 33 - rotulo real 1
Baseline: 1 (100.0%) | NEAT puro: 8 (11.5%) | NEAT + SGD: 1 (34.3%) | similaridade NEAT/SGD: 0.71 | pixels comuns: 93.1%

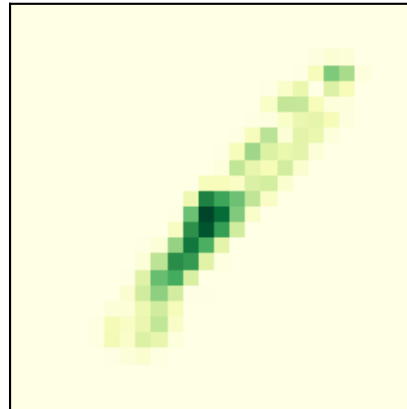
Imagem original



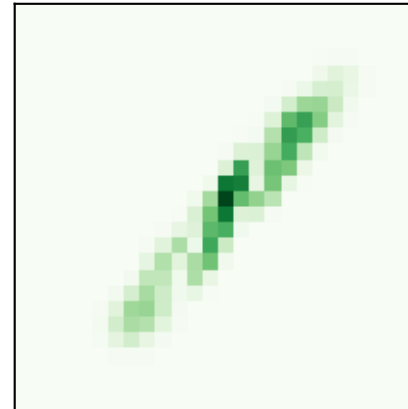
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD



Diferença NEAT vs SGD

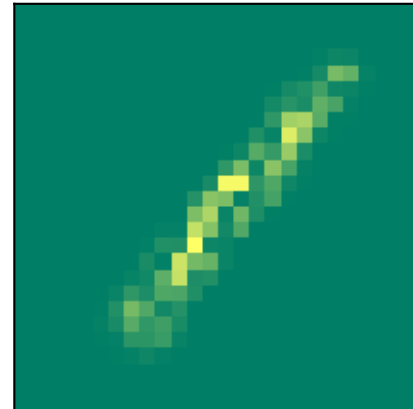
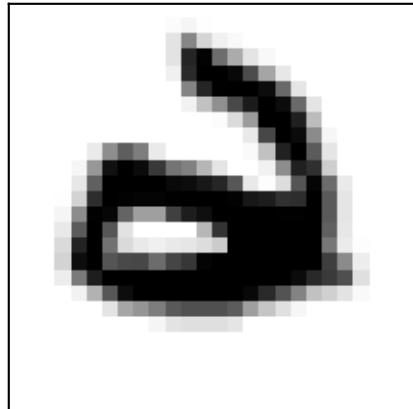
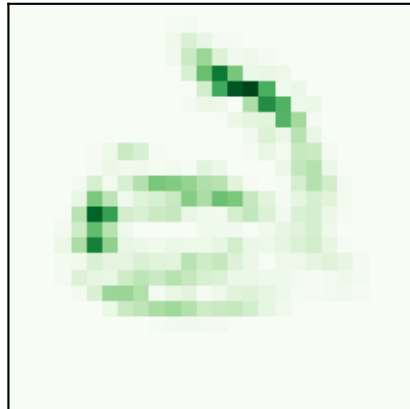


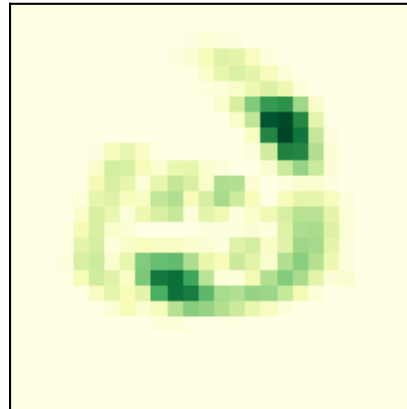
Imagem original



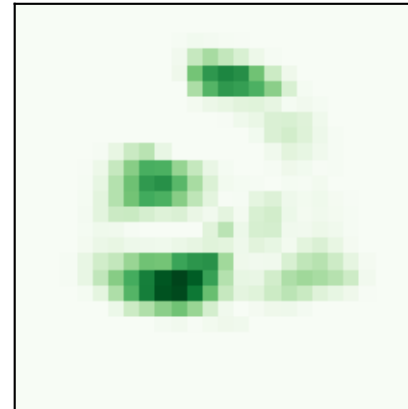
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

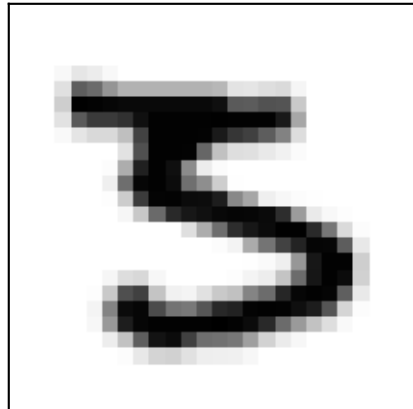


Diferença NEAT vs SGD

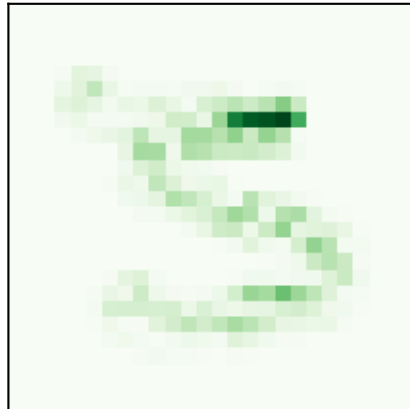


Pagina 10 - discordam - apenas NEAT + SGD acerta - amostra 995 - rotulo real 3
Baseline: 5 (94.1%) | NEAT puro: 5 (11.7%) | NEAT + SGD: 3 (26.1%) | similaridade NEAT/SGD: 0.69 | pixels comuns: 64.4%

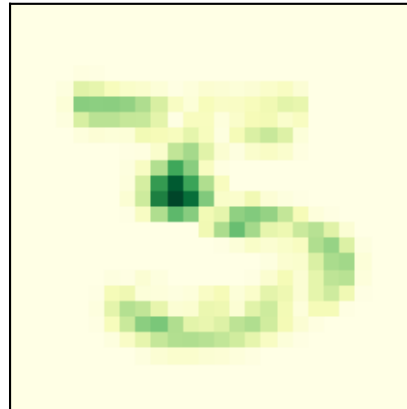
Imagem original



Sensibilidade: baseline



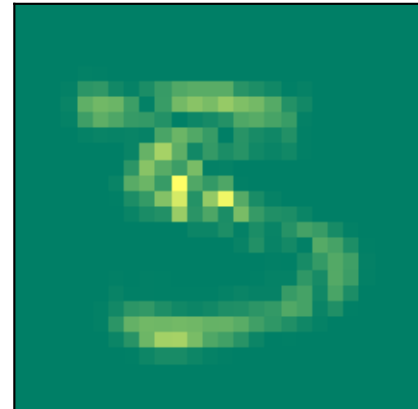
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

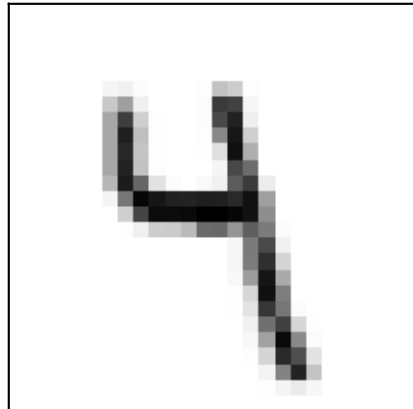


Diferença NEAT vs SGD

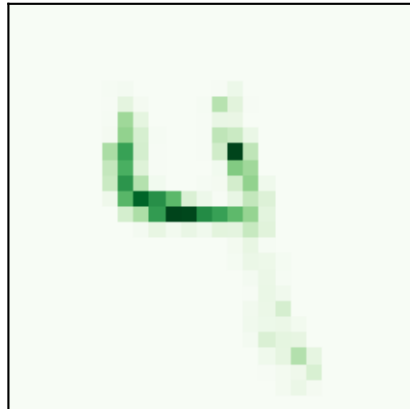


Pagina 11 - discordam - apenas NEAT + SGD acerta - amostra 821 - rotulo real 4
Baseline: 4 (99.9%) | NEAT puro: 9 (11.4%) | NEAT + SGD: 4 (20.9%) | similaridade NEAT/SGD: 0.84 | pixels comuns: 93.1%

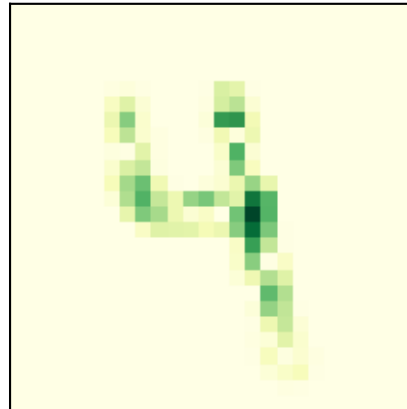
Imagem original



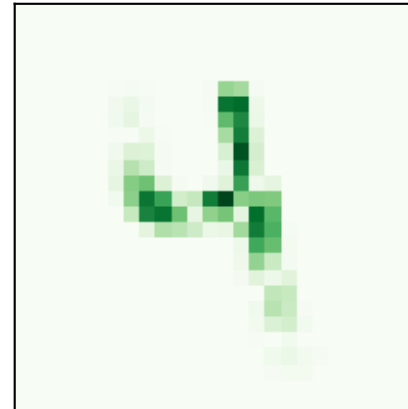
Sensibilidade: baseline



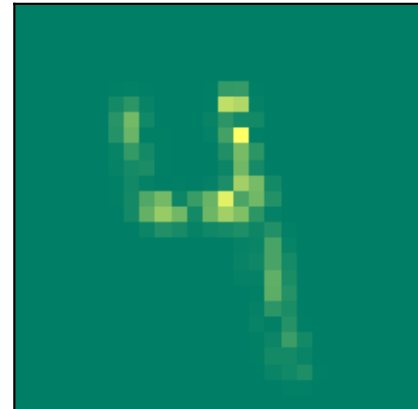
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

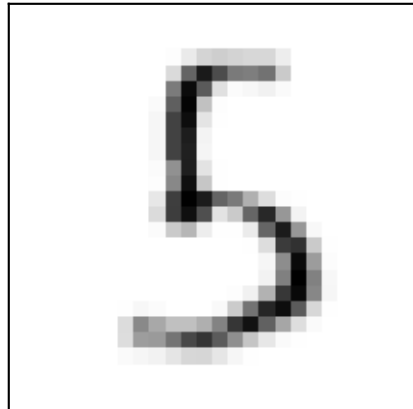


Diferença NEAT vs SGD

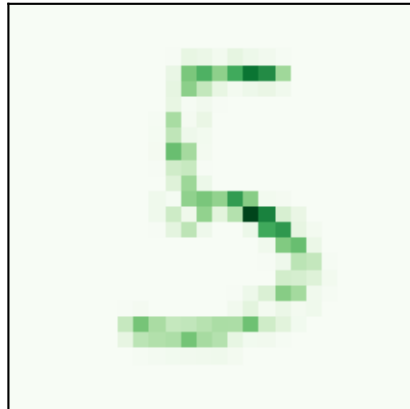


Pagina 12 - discordam - apenas NEAT + SGD acerta - amostra 199 - rotulo real 5
Baseline: 5 (99.8%) | NEAT puro: 3 (12.2%) | NEAT + SGD: 5 (31.6%) | similaridade NEAT/SGD: 0.67 | pixels comuns: 88.1%

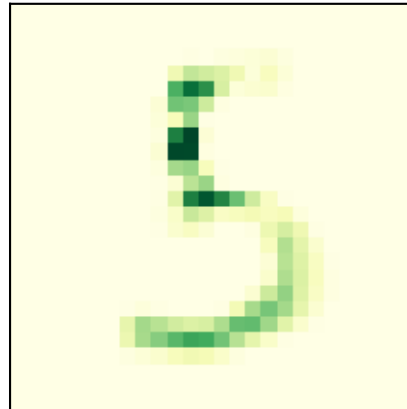
Imagem original



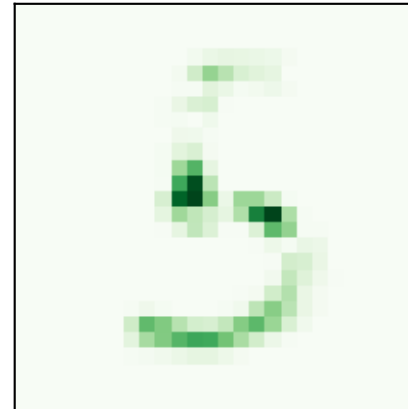
Sensibilidade: baseline



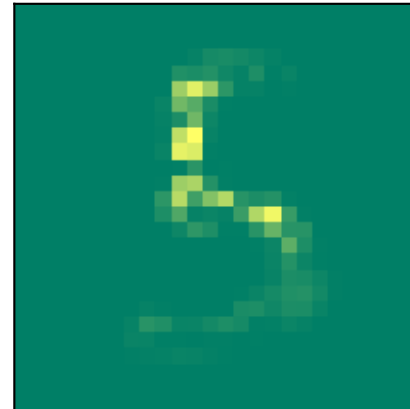
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

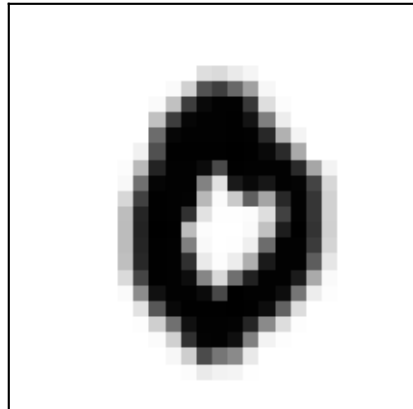


Diferença NEAT vs SGD

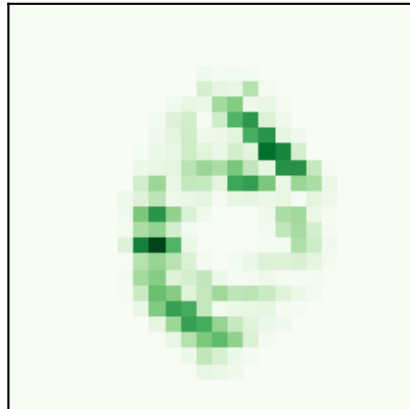


Pagina 13 - discordam - ambos erram classes diferentes - amostra 571 - rotulo real 0
Baseline: 0 (100.0%) | NEAT puro: 6 (11.2%) | NEAT + SGD: 8 (23.5%) | similaridade NEAT/SGD: 0.74 | pixels comuns: 67.3%

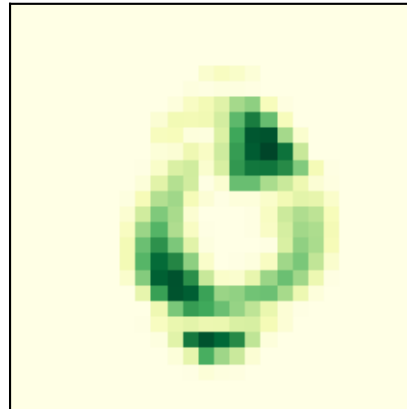
Imagem original



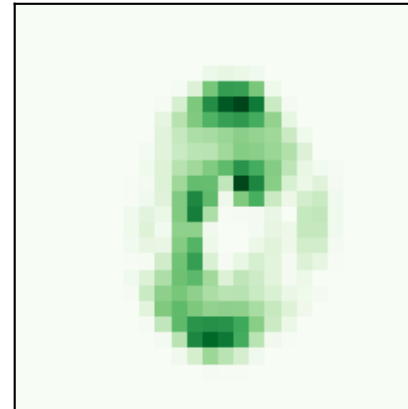
Sensibilidade: baseline



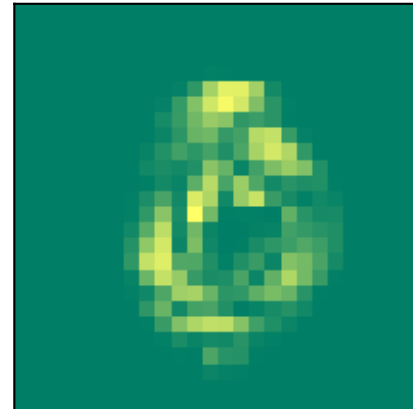
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

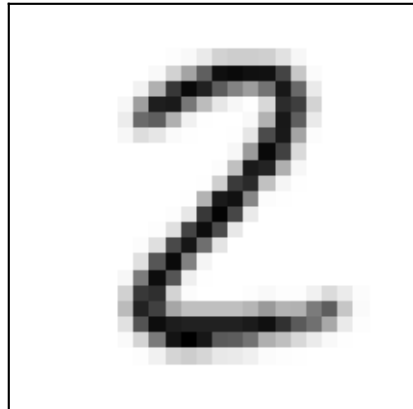


Diferença NEAT vs SGD

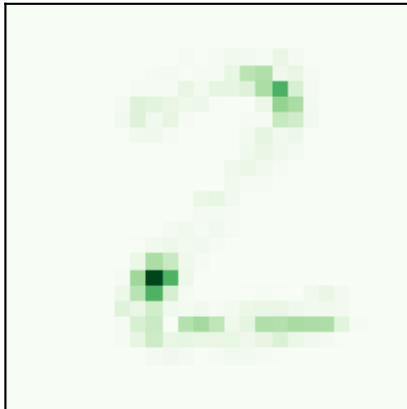


Pagina 14 - discordam - ambos erram classes diferentes - amostra 840 - rotulo real 2
Baseline: 2 (96.8%) | NEAT puro: 8 (11.8%) | NEAT + SGD: 3 (30.0%) | similaridade NEAT/SGD: 0.80 | pixels comuns: 84.2%

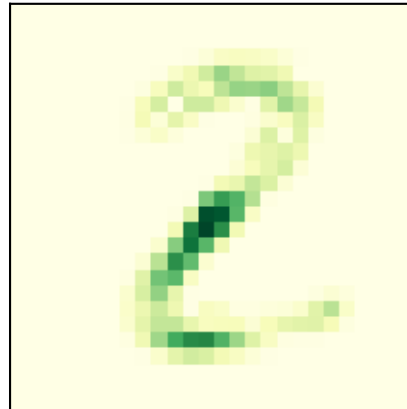
Imagem original



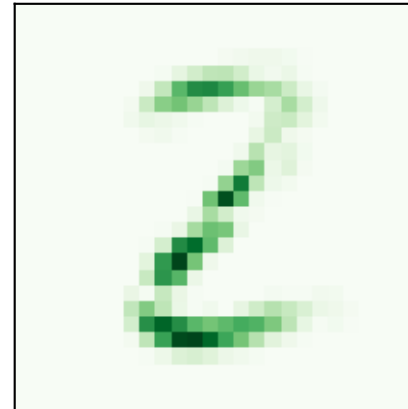
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

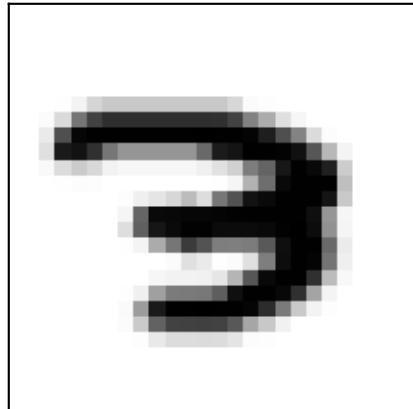


Diferença NEAT vs SGD

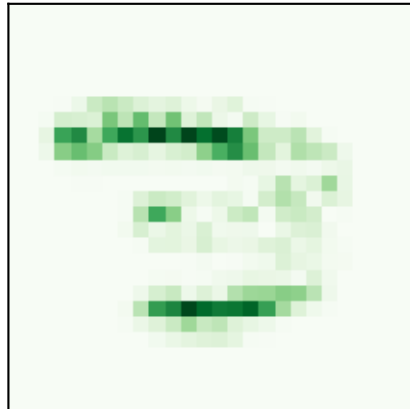


Pagina 15 - discordam - ambos erram classes diferentes - amostra 740 - rotulo real 3
Baseline: 7 (54.6%) | NEAT puro: 7 (10.6%) | NEAT + SGD: 8 (16.0%) | similaridade NEAT/SGD: 0.74 | pixels comuns: 60.4%

Imagem original



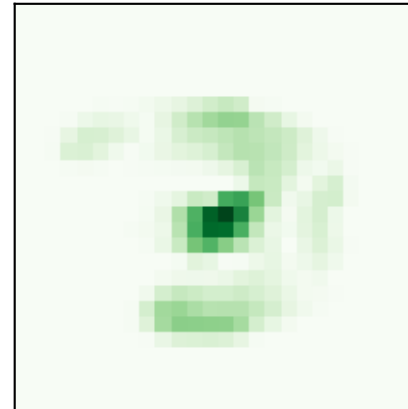
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

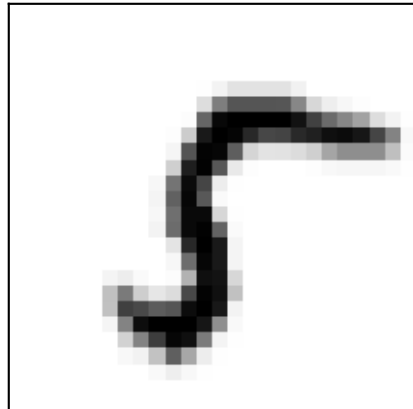


Diferença NEAT vs SGD

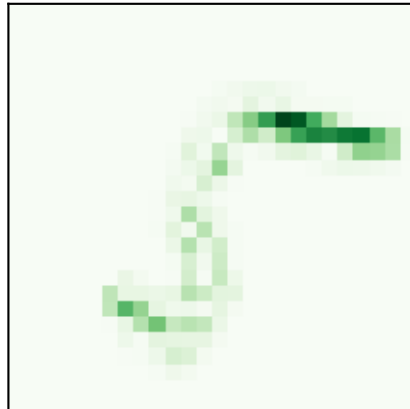


Pagina 16 - discordam - ambos erram classes diferentes - amostra 47 - rotulo real 5
Baseline: 5 (100.0%) | NEAT puro: 1 (11.3%) | NEAT + SGD: 8 (23.1%) | similaridade NEAT/SGD: 0.75 | pixels comuns: 84.2%

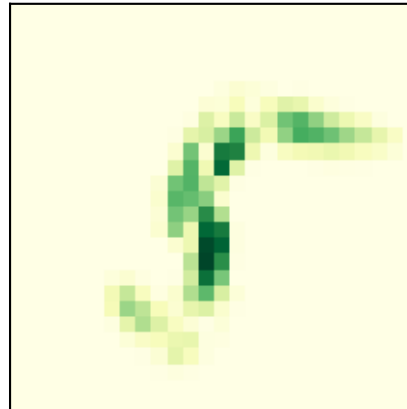
Imagem original



Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

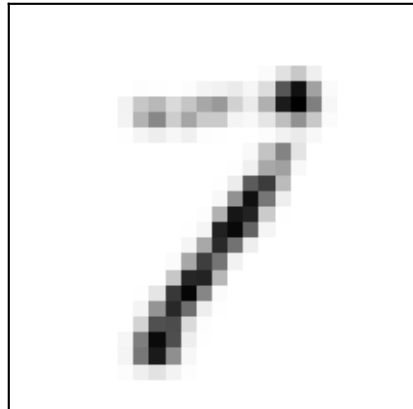


Diferença NEAT vs SGD

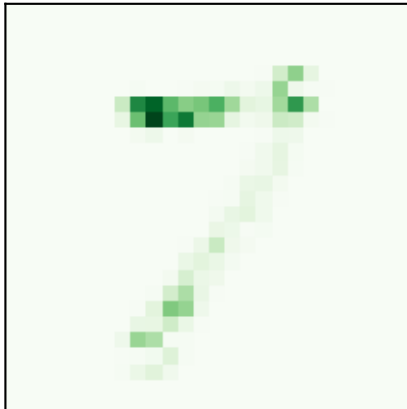


Pagina 17 - discordam - ambos erram classes diferentes - amostra 111 - rotulo real 7
Baseline: 7 (93.8%) | NEAT puro: 8 (11.1%) | NEAT + SGD: 1 (27.7%) | similaridade NEAT/SGD: 0.71 | pixels comuns: 88.1%

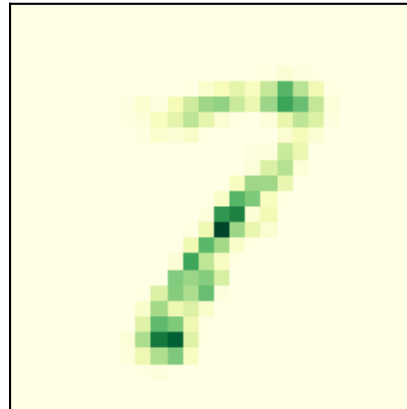
Imagem original



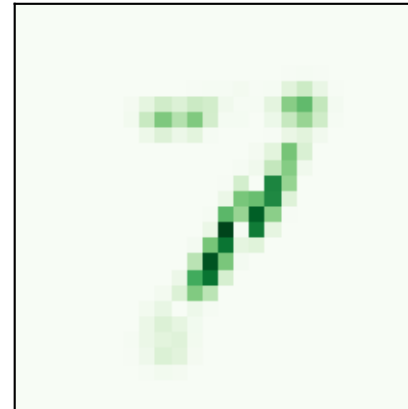
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

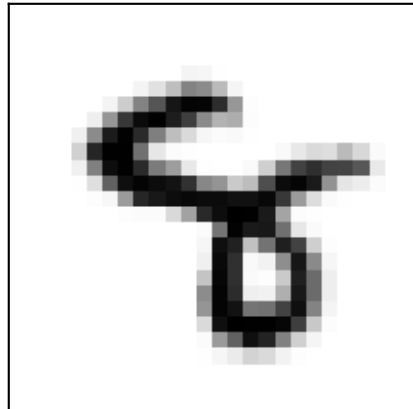


Diferença NEAT vs SGD

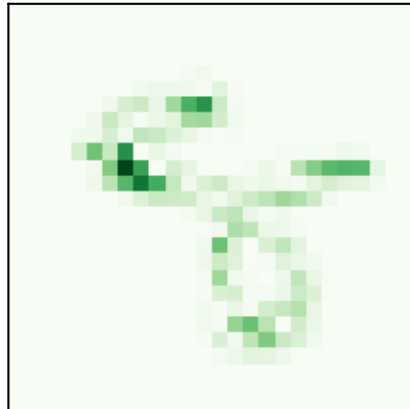


Pagina 18 - discordam - ambos erram classes diferentes - amostra 253 - rotulo real 8
Baseline: 5 (89.3%) | NEAT puro: 9 (10.8%) | NEAT + SGD: 4 (17.7%) | similaridade NEAT/SGD: 0.87 | pixels comuns: 84.2%

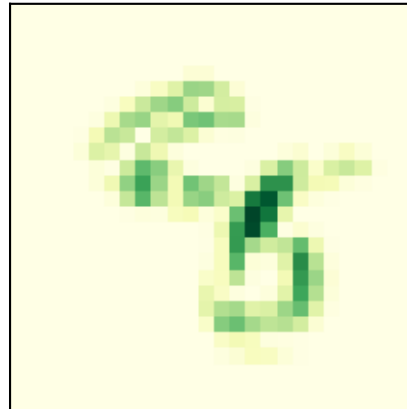
Imagem original



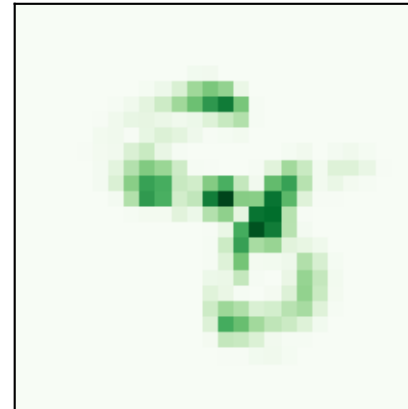
Sensibilidade: baseline



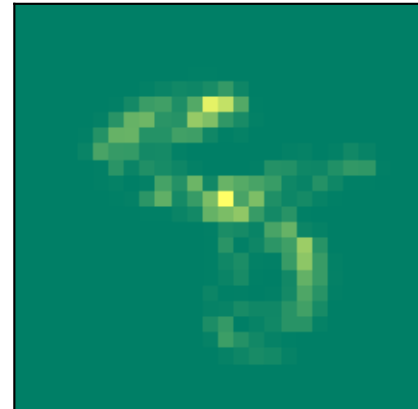
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

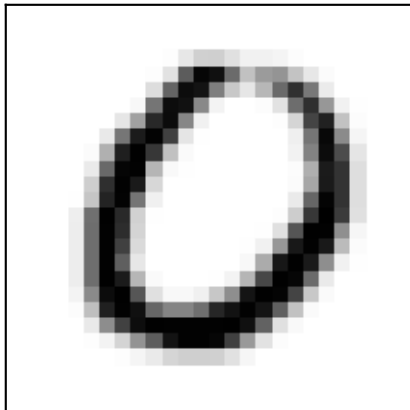


Diferença NEAT vs SGD

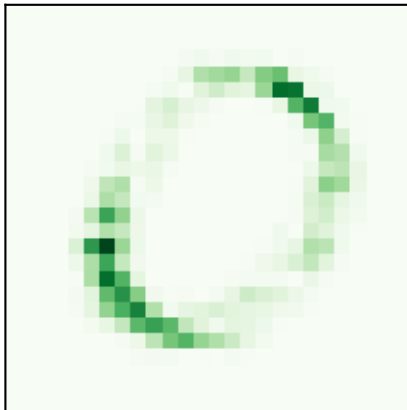


Baseline: 0 (100.0%) | NEAT puro: 0 (14.1%) | NEAT + SGD: 0 (83.3%) | similaridade NEAT/SGD: 0.95 | pixels comuns: 83.2%

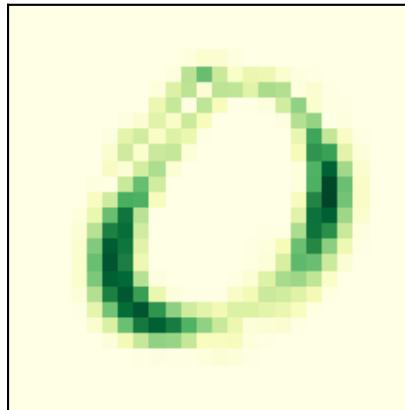
Imagem original



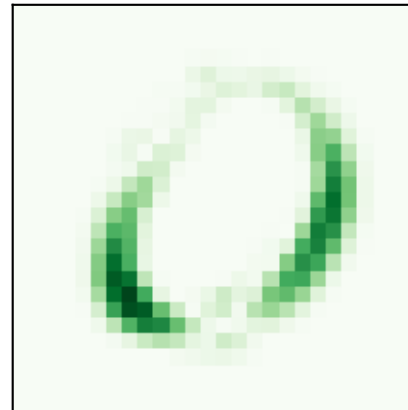
Sensibilidade: baseline



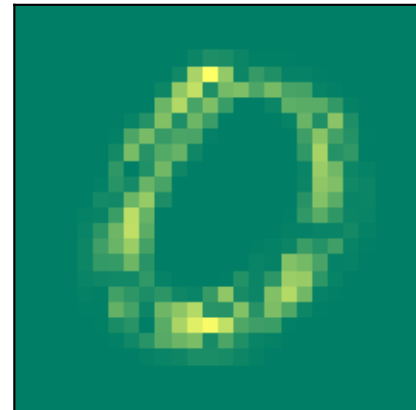
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

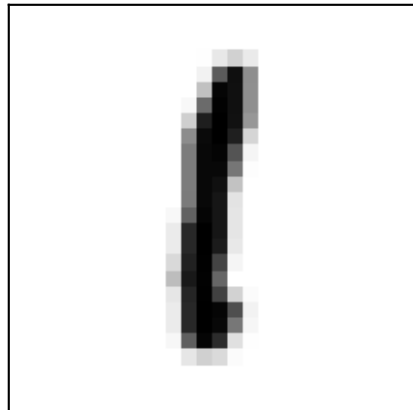


Diferença NEAT vs SGD

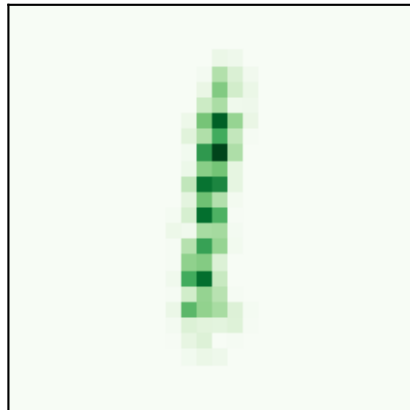


Baseline: 1 (100.0%) | NEAT puro: 1 (14.6%) | NEAT + SGD: 1 (75.8%) | similaridade NEAT/SGD: 0.95 | pixels comuns: 98.0%

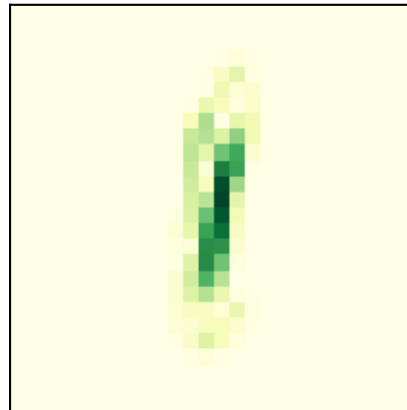
Imagem original



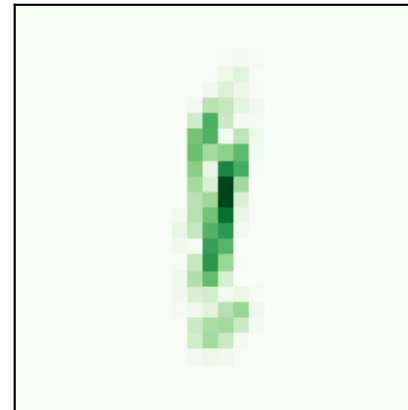
Sensibilidade: baseline



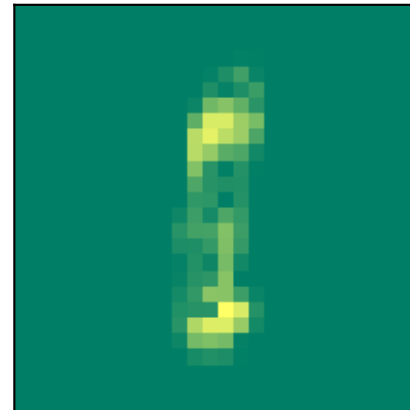
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

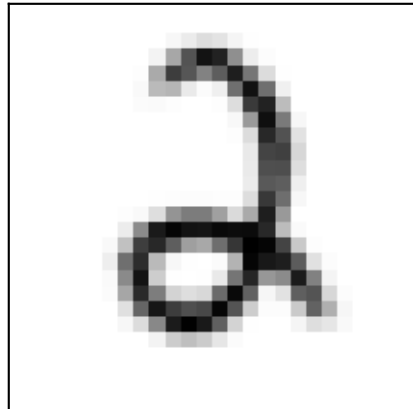


Diferença NEAT vs SGD

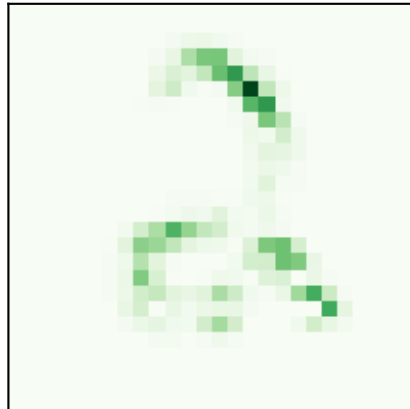


Pagina 21 - concordam - ambos acertam - amostra 24 - rotulo real 2
Baseline: 2 (93.9%) | NEAT puro: 2 (12.2%) | NEAT + SGD: 2 (32.0%) | similaridade NEAT/SGD: 0.90 | pixels comuns: 81.2%

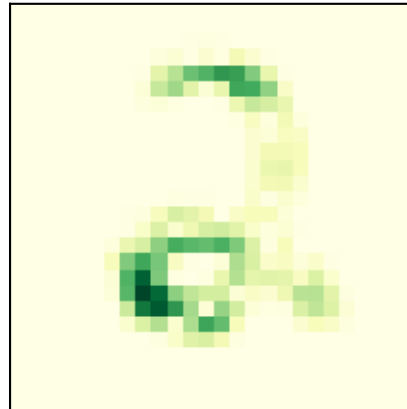
Imagem original



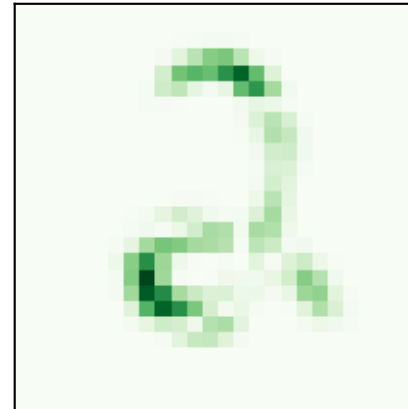
Sensibilidade: baseline



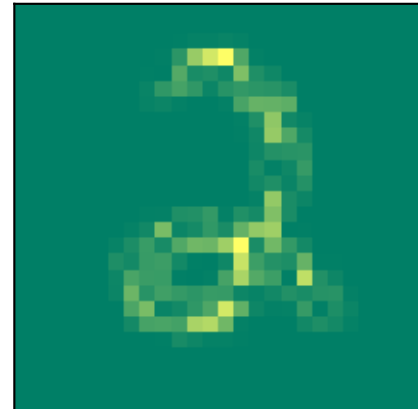
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

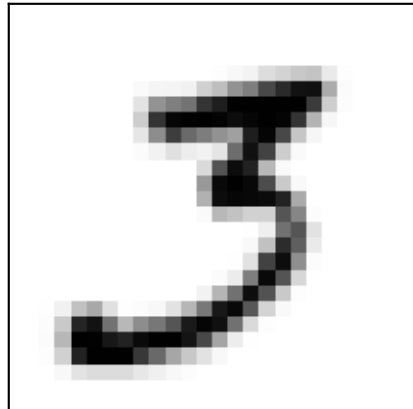


Diferença NEAT vs SGD

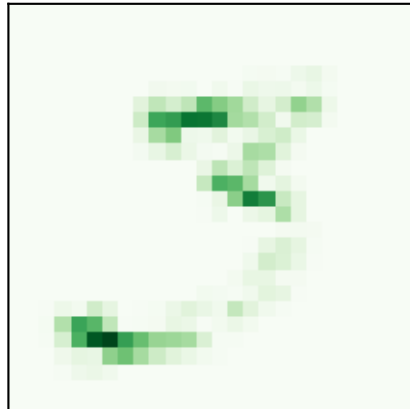


Baseline: 3 (97.8%) | NEAT puro: 3 (12.2%) | NEAT + SGD: 3 (45.2%) | similaridade NEAT/SGD: 0.90 | pixels comuns: 79.2%

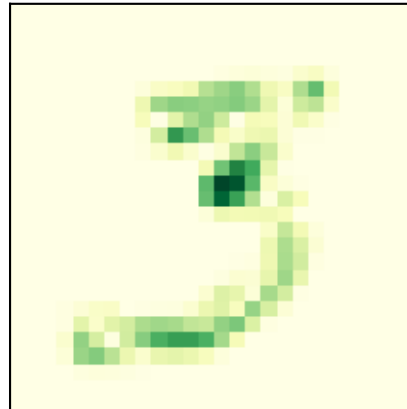
Imagem original



Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

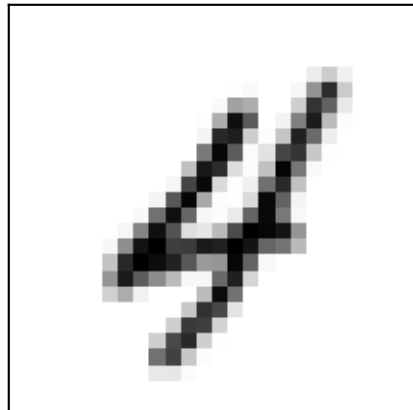


Diferença NEAT vs SGD

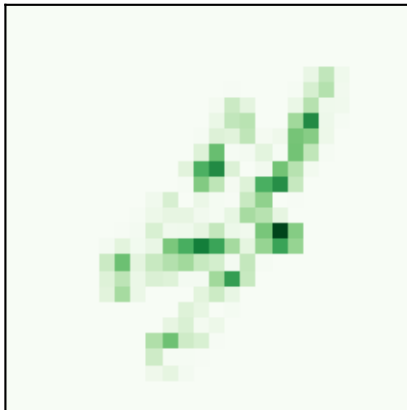


Baseline: 4 (99.9%) | NEAT puro: 4 (11.2%) | NEAT + SGD: 4 (22.7%) | similaridade NEAT/SGD: 0.97 | pixels comuns: 93.1%

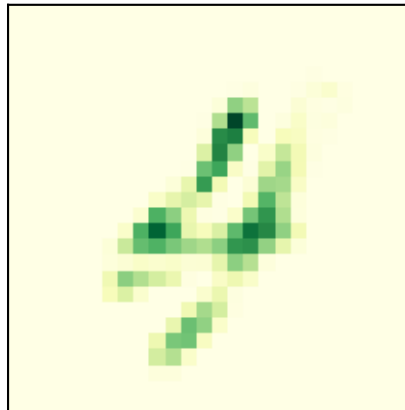
Imagem original



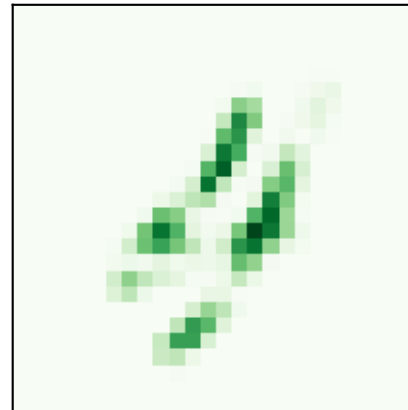
Sensibilidade: baseline



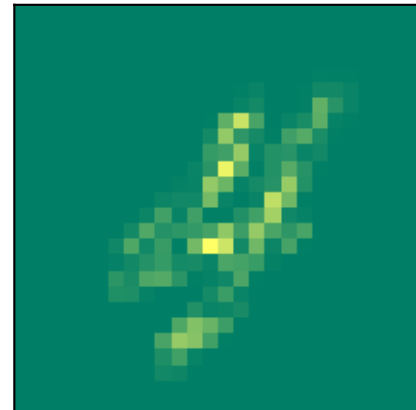
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

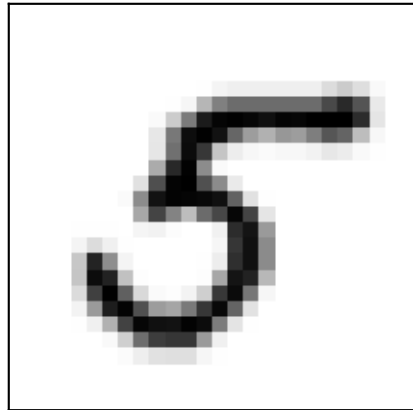


Diferença NEAT vs SGD

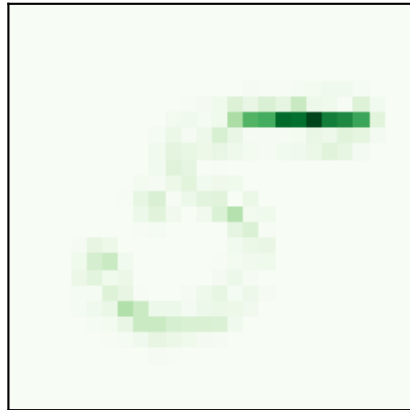


Pagina 24 - concordam - ambos acertam - amostra 44 - rotulo real 5
Baseline: 5 (100.0%) | NEAT puro: 5 (12.3%) | NEAT + SGD: 5 (26.0%) | similaridade NEAT/SGD: 0.89 | pixels comuns: 76.2%

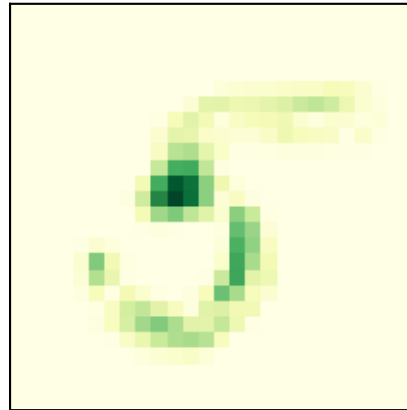
Imagem original



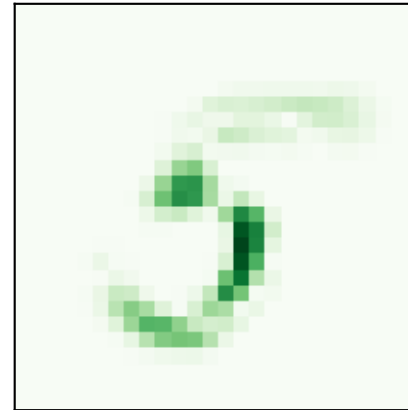
Sensibilidade: baseline



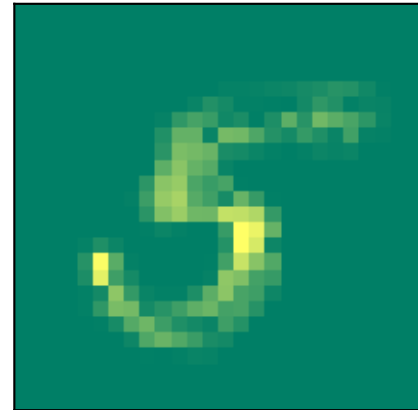
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

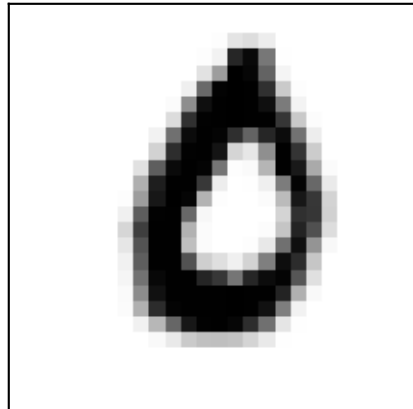


Diferença NEAT vs SGD

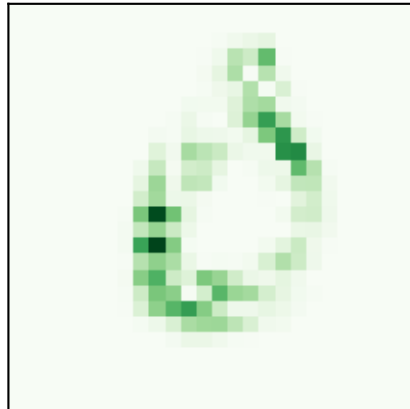


Pagina 25 - concordam - ambos erram a mesma classe - amostra 10 - rotulo real 0
Baseline: 0 (100.0%) | NEAT puro: 6 (11.7%) | NEAT + SGD: 6 (25.1%) | similaridade NEAT/SGD: 0.98 | pixels comuns: 92.1%

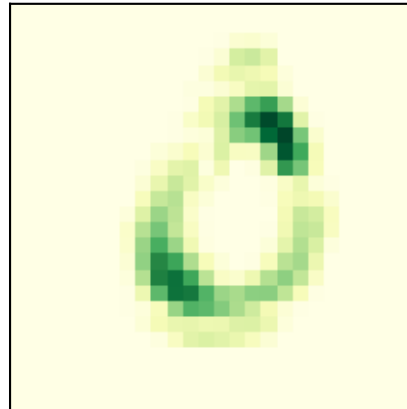
Imagem original



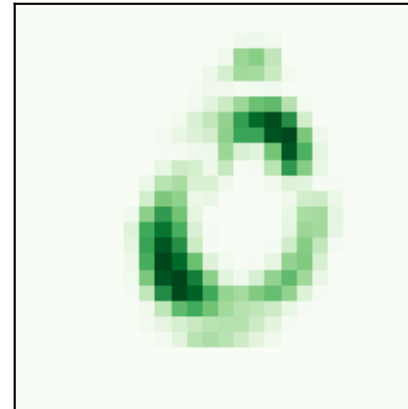
Sensibilidade: baseline



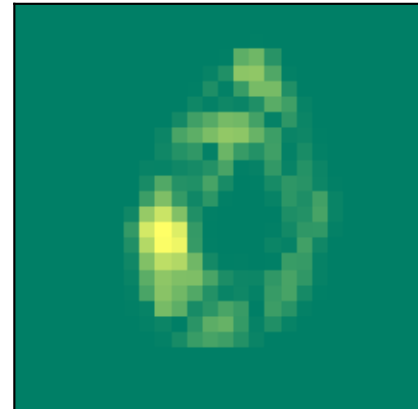
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

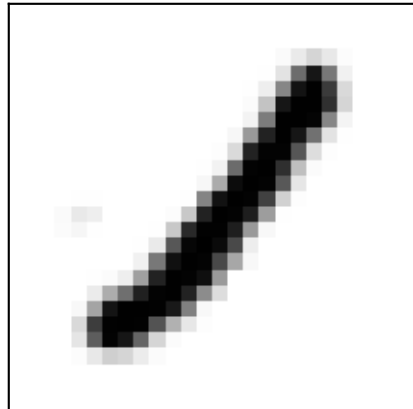


Diferença NEAT vs SGD

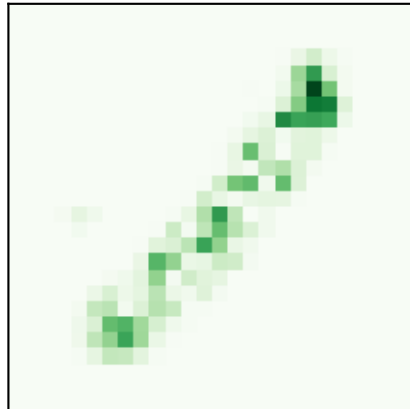


Baseline: 1 (100.0%) | NEAT puro: 2 (11.6%) | NEAT + SGD: 2 (30.6%) | similaridade NEAT/SGD: 0.92 | pixels comuns: 90.1%

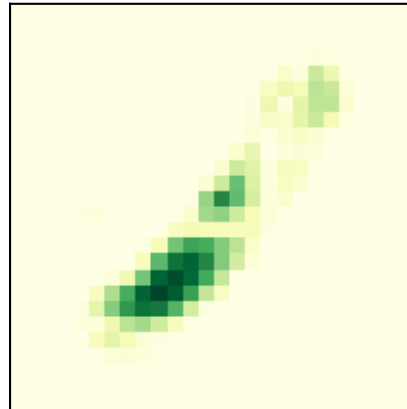
Imagem original



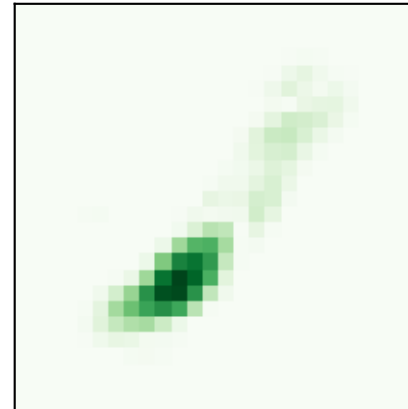
Sensibilidade: baseline



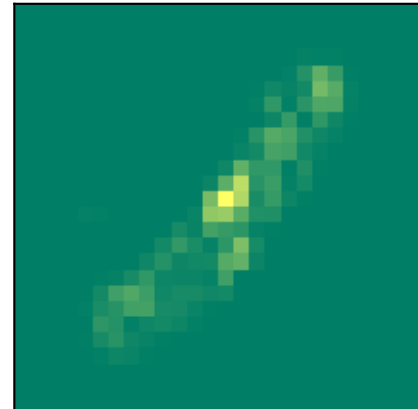
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

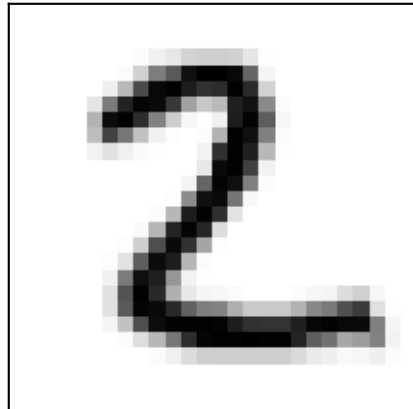


Diferença NEAT vs SGD

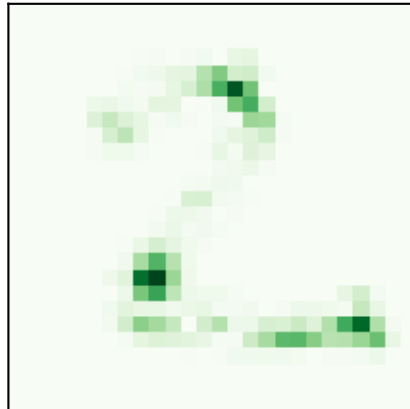


Pagina 27 - concordam - ambos erram a mesma classe - amostra 11 - rotulo real 2
Baseline: 2 (99.8%) | NEAT puro: 3 (11.7%) | NEAT + SGD: 3 (34.7%) | similaridade NEAT/SGD: 0.93 | pixels comuns: 86.1%

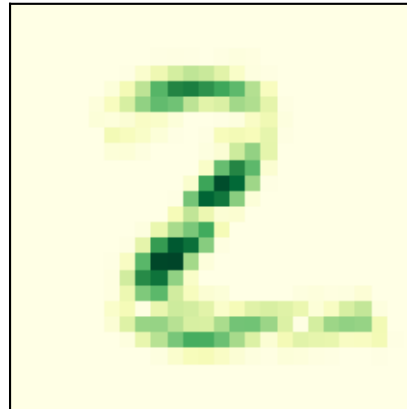
Imagem original



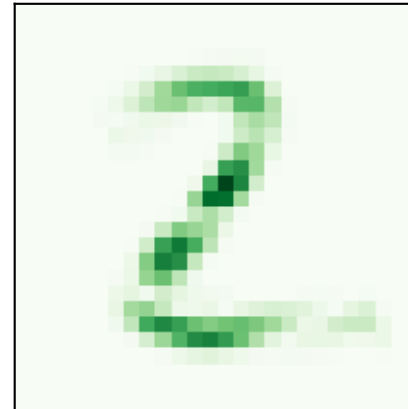
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

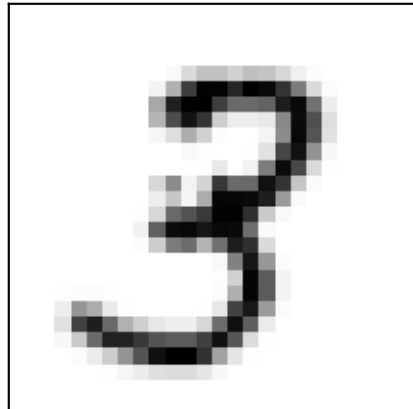


Diferença NEAT vs SGD

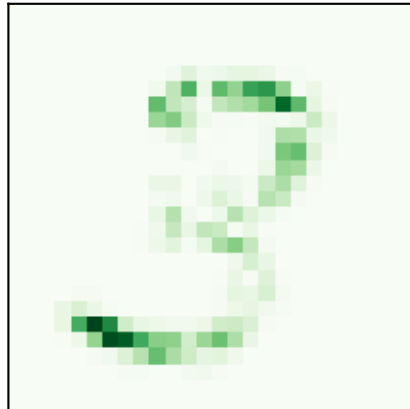


Pagina 28 - concordam - ambos erram a mesma classe - amostra 23 - rotulo real 3
Baseline: 3 (94.6%) | NEAT puro: 8 (12.0%) | NEAT + SGD: 8 (30.6%) | similaridade NEAT/SGD: 0.93 | pixels comuns: 80.2%

Imagem original



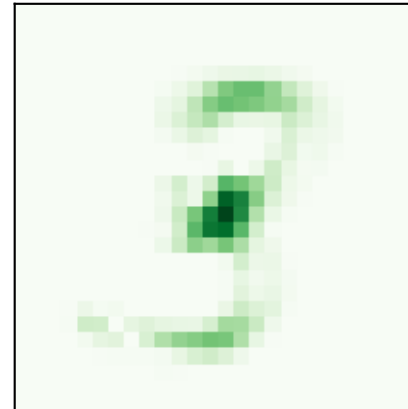
Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

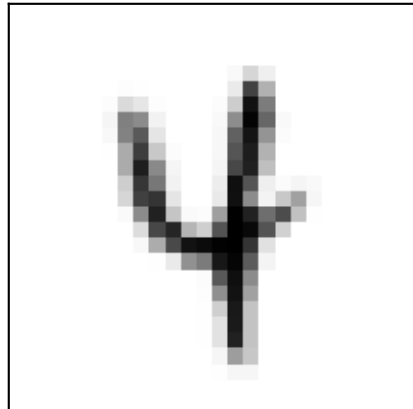


Diferença NEAT vs SGD

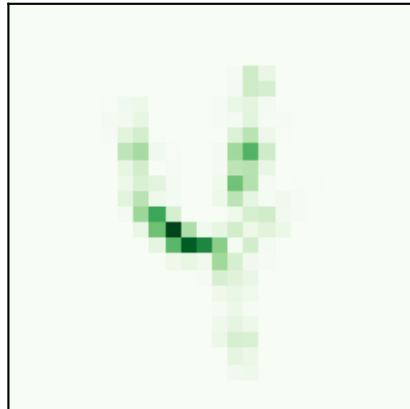


Pagina 29 - concordam - ambos erram a mesma classe - amostra 16 - rotulo real 4
Baseline: 4 (99.9%) | NEAT puro: 9 (12.0%) | NEAT + SGD: 9 (32.2%) | similaridade NEAT/SGD: 0.93 | pixels comuns: 89.1%

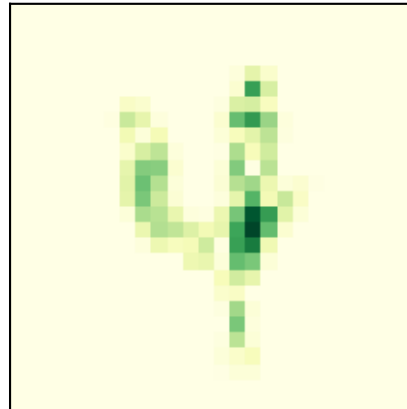
Imagem original



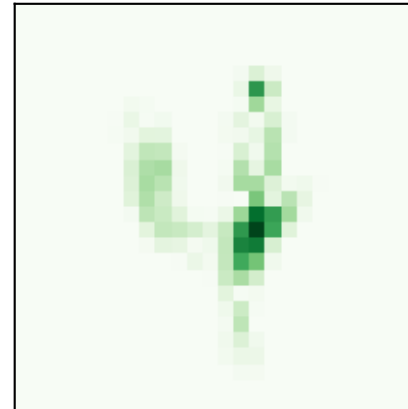
Sensibilidade: baseline



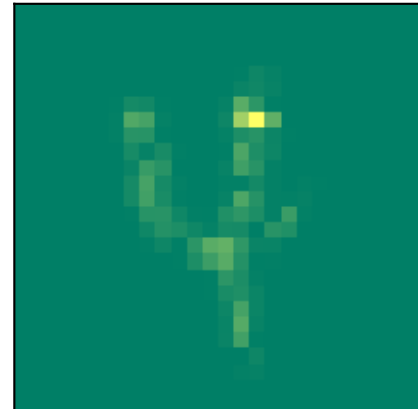
Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD

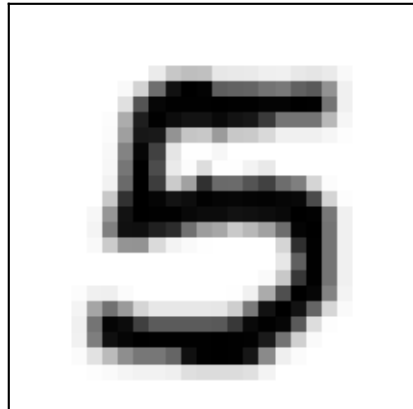


Diferença NEAT vs SGD



Pagina 30 - concordam - ambos erram a mesma classe - amostra 78 - rotulo real 5
Baseline: 5 (100.0%) | NEAT puro: 3 (11.5%) | NEAT + SGD: 3 (27.0%) | similaridade NEAT/SGD: 0.93 | pixels comuns: 80.2%

Imagem original



Sensibilidade: baseline



Sensibilidade: NEAT puro



Sensibilidade: NEAT + SGD



Diferença NEAT vs SGD

